

LECTURE NOTES MATH255

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1. LECTURE 13

In this class we investigate algebraic extensions and algebraic closures.

1.1. Algebraic extensions. If F is a field, then $F \leq E$ is an algebraic extension of F if every element of E is algebraic over F :

Definition 1.1. Let F be a field. An extension $F \leq E$ of F is an algebraic extension if every $\alpha \in E$ is algebraic over F . That is, E is an algebraic extension of F if for every $\alpha \in E$ there is a polynomial $f(x) \in F[x]$ such that $f(\alpha) = 0$.

We also introduce the related notion of a finite extension:

Definition 1.2. Let F be a field. An extension $F \leq E$ of F is a finite extension if the dimension of E over F is finite. We write this dimension as $[E : F]$ ($[E : F]$ is usually called the degree of the extension).

Example 1.1. Let $F = \mathbb{Q}$ and $E = \mathbb{Q}(\sqrt{7})$. Any element of E can be written as $\alpha = a + b\sqrt{7}$, $a, b \in \mathbb{Q}$. Note that if we write:

$$(1.1) \quad (\alpha - a)^2 = 7b^2, \quad 0 = (\alpha - a)^2 - 7b^2$$

so $\alpha = a + b\sqrt{7}$ is a zero of the polynomial $g(x) = (x - a)^2 - 7b^2 \in \mathbb{Q}[x]$. That is, $\mathbb{Q}(\sqrt{7})$ is an algebraic extension of $\mathbb{Q}$.

Example 1.2. We can generalize the previous example significantly. In the homework we prove that if $F \leq E$ is a finite extension, then every $\alpha \in E$ is algebraic over F . That is, any finite extension $F \leq E$ is an algebraic extension.

Example 1.3. The converse of the previous example does not hold. That is, there are algebraic extensions that are not finite. Indeed, if we define:

$$(1.2) \quad \overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}$$

then $\overline{\mathbb{Q}}$ is a field (see Theorem 1.6). By construction we see that $\overline{\mathbb{Q}}$ is an algebraic extension of $\mathbb{Q}$, however it is not a finite extension. Indeed, for any $n \in \{2, 3, 4, 5, 6, \dots\}$ we define a polynomial:

$$(1.3) \quad f_n(x) = x^n + 2x^{n-1} + 2x^{n-2} + \dots + 2x^2 + 2x + 2.$$

Noting that the leading coefficient of $f_n(x)$ is not divisible by 2, but every other coefficient is divisible by 2, and the constant term is not divisible by $2^2 = 4$ it follows

that $f_n(x)$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion. Let α_n be a root of $f_n(x)$, then $f_n(x)$ is the irreducible polynomial for α_n so $\deg(\alpha_n) = \deg(f_n(x)) = n$. It follows that the dimension of $\mathbb{Q}(\alpha_n)$ over $\mathbb{Q}$ is n . On the other hand, as we saw in the previous class, every element of $\mathbb{Q}(\alpha_n)$ is algebraic over $\mathbb{Q}$ so $\mathbb{Q}(\alpha_n)$ is an algebraic extension of $\mathbb{Q}$. It follows that $\mathbb{Q}(\alpha_n) \leq \overline{\mathbb{Q}}$. Since $\overline{\mathbb{Q}}$, viewed as a vector space over $\mathbb{Q}$, contains a subspace of dimension n for every $n \geq 2$ it follows that $\overline{\mathbb{Q}}$ is not a finite dimensional vector space over $\mathbb{Q}$, and therefore not a finite extension of $\mathbb{Q}$.

In the homework we prove the following two theorems on finite extensions:

Theorem 1.1. *Any finite extension $F \leq E$ is an algebraic extension.*

Theorem 1.2. *If $F \leq E$ and $E \leq L$ are finite extensions then $F \leq L$ is a finite extension and we have the formula $[L : F] = [E : F] \cdot [L : E]$.*

An immediate consequence of this theorem is the following theorem which is sometimes useful when studying field extensions.

Theorem 1.3. *Let E be a finite extension of F and let $\alpha \in E$ be algebraic over F . Then $\deg(\alpha)$ divides $[E : F]$.*

Proof. Let $F(\alpha)$ be the simple extension of α (i.e., the smallest subfield of E that contains α and F). Then we have $F \leq F(\alpha) \leq E$ and so: $[E : F] = [F(\alpha) : F] \cdot [E : F(\alpha)]$, but $\deg(\alpha) = [F(\alpha) : F]$ so we see that $\deg(\alpha)$ divides $[E : F]$. ■

Example 1.4. The number $\sqrt{2}$ is not contained in $\mathbb{Q}(2^{1/3})$. Indeed, note that $2^{1/3}$ is a root of $f(x) = x^3 - 2$. Moreover, the roots of $f(x)$ are $2^{1/3}, 2^{1/3}e^{2\pi i/3}, 2^{1/3}e^{4\pi i/3}$ none of which are rational, so since $f(x)$ has degree 3 we conclude that $f(x)$ is irreducible. It follows that $f(x) = x^3 - 2$ is the irreducible polynomial for $2^{1/3}$, so $\deg(2^{1/3}) = [\mathbb{Q}(2^{1/3}) : \mathbb{Q}] = 3$. It follows that any element $\beta \in \mathbb{Q}(2^{1/3})$ must have degree 1 (which implies that it is rational) or 3. In particular, $\sqrt{2}$ has degree 2 (since its irreducible polynomial is $x^2 - 2$), so $\sqrt{2} \notin \mathbb{Q}(2^{1/3})$.

Definition 1.3. *Let $F \leq E$ be a field extension and $\alpha_1, \dots, \alpha_n \in E$ be algebraic over F . Then we define $F(\alpha_1, \dots, \alpha_n)$ as the smallest subfield of E that contains F and $\alpha_1, \dots, \alpha_n$.*

Theorem 1.4. *Let $E \leq F$ be an algebraic extension of F . Then E is a finite extension if and only if $E = F(\alpha_1, \dots, \alpha_n)$ for some elements $\alpha_1, \dots, \alpha_n \in E$.*

Proof. That $F(\alpha_1, \dots, \alpha_n)$ is a finite extension of F is proved in the homework, so if $E = F(\alpha_1, \dots, \alpha_n)$ then E is a finite extension of F .

Conversely, let E be a finite extension of F . Let $\beta_1, \dots, \beta_n \in E$ be a F -basis of E . Then we claim that $F(\beta_1, \dots, \beta_n) = E$. Indeed, it is clear that $F(\beta_1, \dots, \beta_n) \subset E$ since $F \subset E$ and $\beta_1, \dots, \beta_n \in E$. However, $F(\beta_1, \dots, \beta_n)$ is a field that contains F and $\beta_1, \dots, \beta_n$ so $F(\beta_1, \dots, \beta_n)$ contains any element of the form:

$$(1.4) \quad a_1\beta_1 + \dots + a_n\beta_n \in F(\beta_1, \dots, \beta_n), \quad a_1, \dots, a_n \in F.$$

Since $\beta_1, \dots, \beta_n$ is a basis for E as a F -vector space we know that any $\alpha \in E$ can be written as $\alpha = a_1\beta_1 + \dots + a_n\beta_n$ for some $a_1, \dots, a_n \in F$, so we also get the reverse inclusion $E \subset F(\beta_1, \dots, \beta_n)$. Combined we conclude that $E = F(\beta_1, \dots, \beta_n)$. ■

1.2. Algebraic closures. Let $F \leq E$ be a field extension. We define the algebraic closure of F in E as:

$$(1.5) \quad \overline{F}_E = \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}.$$

We have the following important result:

Theorem 1.5. *Let $F \leq E$ be a field extension. Then the algebraic closure of F in E , $\overline{F}_E$, is a subfield of E .*

Proof. We begin by proving that if $0 \neq \alpha \in \overline{F}_E$ then $1/\alpha \in \overline{F}_E$. To see this, note that $F(\alpha) \subset \overline{F}_E$ since any element in $F(\alpha)$ is algebraic if α is algebraic. But $F(\alpha)$ is a field, so $1/\alpha \in F(\alpha) \subset \overline{F}_E$.

It remains to show that $\overline{F}_E$ is closed under addition and multiplication, and that $1 \in \overline{F}_E$. To see that $1 \in \overline{F}_E$ we simply note that 1 is a zero of $x - 1 \in F[x]$. Finally, let $\alpha, \beta \in \overline{F}_E$ and consider the field $F(\alpha, \beta)$. By Theorem 1.4 we know that $F(\alpha, \beta)$ is a finite extension and since any finite extension is algebraic we conclude that $F(\alpha, \beta) \subset \overline{F}_E$. But $\alpha + \beta, \alpha\beta \in F(\alpha, \beta)$ which implies that $\alpha + \beta, \alpha\beta \in \overline{F}_E$. ■

Theorem 1.6. *The set of algebraic numbers*

$$(1.6) \quad \{\alpha \in \mathbb{C} \mid f(\alpha) = 0, \text{ for some non-constant } f(x) \in \mathbb{Q}[x]\} = \overline{\mathbb{Q}} \subset \mathbb{C},$$

form a field.

Proof. The theorem follows from the previous theorem once we note that $\overline{\mathbb{Q}} = \overline{\mathbb{Q}}_{\mathbb{C}}$. ■

Definition 1.4. *A field F is algebraically closed if every non-zero polynomial $f(x) \in F[x]$ has a zero in F .*

Example 1.5. The field $\mathbb{C}$ is algebraically closed by the fundamental theorem of algebra. We will not prove this in this course, but for those familiar with complex analysis, it is a consequence of Liouville's theorem. Indeed, if $p(x) \in \mathbb{C}[x]$ has no zeros then $g(x) = 1/p(x)$ is a bounded holomorphic function. But by Liouville's theorem any bounded holomorphic function is constant, so $g(x)$ is constant. But if $1/p(x)$ is constant, then $p(x)$ is also constant, so any polynomial without a zero in $\mathbb{C}$ is necessarily constant.

Theorem 1.7. *If F is algebraically closed and $f(x) \in F[x]$ then $f(x)$ can be decomposed into linear factors in $F[x]$: there are $a_1, \dots, a_n, c \in F$ such that $f(x) = c(x - a_1)\dots(x - a_n)$.*

Proof. Let F be algebraically closed and $f(x) \in F[x]$. If $f(x)$ is constant then there is nothing to prove, so assume that $\deg(f(x)) \geq 1$. Since $f(x)$ is non-constant we find a zero a_1 of $f(x)$, and write $f(x) = (x - a_1)g(x)$. If $g(x) = c \in F$ is constant then we are done. If $g(x)$ is non-constant then we can write $g(x) = (x - a_2)h(x)$ for some zero $a_2 \in F$ of $g(x)$. So we have $f(x) = (x - a_1)(x - a_2)h(x)$. Proceeding by induction the theorem follows. ■

1.3. Algebraic closure of a field. Let F be a field. An algebraic closure of F is an algebraic extension of F that is algebraically closed, we write this as a definition.

Definition 1.5. Let F be a field. An algebraic closure $\overline{F}$ of F is an algebraic extension of F such that $\overline{F}$ is algebraically closed.

Theorem 1.8. The algebraic numbers $\overline{\mathbb{Q}}$ is an algebraic closure of $\mathbb{Q}$.

Proof. Note that $\overline{\mathbb{Q}}$ is an algebraic extension of $\mathbb{Q}$. Indeed, $\overline{\mathbb{Q}}$ is a field by Theorem 1.6, and it is an extension of $\mathbb{Q}$ since for any $q \in \mathbb{Q}$ it holds that q is a zero of $x - q \in \mathbb{Q}[x]$, so $\mathbb{Q} \leq \overline{\mathbb{Q}}$. It remains to show that $\overline{\mathbb{Q}}$ is algebraically closed. To see this, let $f(x) \in \overline{\mathbb{Q}}[x]$ with coefficients $f(x) = c_0 + c_1x + \dots + c_nx^n$ with $c_0, \dots, c_n \in \overline{\mathbb{Q}}$.

Let $\alpha \in \mathbb{C}$ be a zero of $f(x)$. By Theorem 1.4 we know that $E = \mathbb{Q}(c_0, \dots, c_n)$ is a finite extension of $\mathbb{Q}$ since each $c_0, \dots, c_n$ is algebraic over $\mathbb{Q}$. Moreover, $E(\alpha)$ is a finite extension of E since α is algebraic over E (note that $f(x) \in E[x]$). It follows by Theorem 1.2 that $E(\alpha)$ is a finite extension of $\mathbb{Q}$. Since any finite extension of $\mathbb{Q}$ is algebraic, we conclude that $E(\alpha)$ is an algebraic extension of $\mathbb{Q}$. Finally, note that $\alpha \in E(\alpha)$ so since $E(\alpha)$ is an algebraic extension of $\mathbb{Q}$ we conclude that there is some non-constant polynomial $g(x) \in \mathbb{Q}[x]$ such that $g(\alpha) = 0$. That is, α is a root of a polynomial with coefficients in $\mathbb{Q}$, so $\alpha \in \overline{\mathbb{Q}}$. That is, if we take any non-constant polynomial $f(x) \in \overline{\mathbb{Q}}[x]$ then $f(x)$ has a root in $\overline{\mathbb{Q}}$ so $\overline{\mathbb{Q}}$ is algebraically closed. ■

Notice that we produced an algebraic closure of $\mathbb{Q}$ by roughly speaking taking the set of all zeros of polynomials in $\mathbb{Q}[x]$. This procedure largely works for any field F , but it becomes more complicated because we usually do not have an algebraically closed field $\mathbb{C}$ in which we can produce all zeros first. Nonetheless, we have the following theorem.

Theorem 1.9. Every field F has an algebraic closure $\overline{F}$.

To prove this theorem we need to introduce partial orderings and Zorn's lemma.

Definition 1.6. Let S be a set. A relation $\leq$ is a partial ordering on S if it satisfies:

- (i) $a \leq a$ for all $a \in S$,
- (ii) if $a \leq b$ and $b \leq a$ then $a = b$,
- (iii) if $a \leq b$ and $b \leq c$ then $a \leq c$.

Remark 1. Note that not all elements in S must be related by $\leq$. That is, there might be $a, b \in S$ such that neither $a \leq b$ or $b \leq a$ hold. A good example to keep in mind is the relation of less than or equal to between functions on $\mathbb{R}$. If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are functions then $f \leq g$ if $f(x) \leq g(x)$ for all $x \in \mathbb{R}$. This defines a partial ordering, but we can not compare all elements with this ordering. Indeed, if $f(x) = x^2$ and $g(x) = x^3$ then $g(x) > f(x)$ for $x > 1$ but $g(x) < f(x)$ for $x < 0$, so we do not have $f \leq g$ or $g \leq f$ (in this case we simply say that f and g are not comparable).

Definition 1.7. Let S be a set with a partial ordering $\leq$. We say that $T \subset S$ is a chain if every $a, b \in T$ are comparable (i.e., for every $a, b \in T$ we either have $a \leq b$ or $b \leq a$).

Definition 1.8. Let S be a set with a partial ordering. An element $u \in S$ is an upper bound for a subset $A \subset S$ if $a \leq u$ for all $a \in A$. An element $m \in S$ is maximal if $m \leq s$ implies $m = s$ (i.e., there is no element that is larger than m with respect to the partial ordering).

Theorem 1.10 (Zorn's lemma). Let S be a set with a partial order $\leq$. If every chain $T \subset S$ has an upper bound u , then S has a maximal element $m \in S$.

Remark 2. The upper bound might not be unique.

Remark 3. We will not prove this theorem, but it is a consequence of the axiom of choice.

We can now prove Theorem 1.9.

Proof of Theorem 1.9. Let F be a field and define:

$$(1.7) \quad S = \{F \leq E \mid E \text{ is an algebraic extension of } F\}.$$

We introduce a partial ordering on S by $E \leq L$ if L is an extension of E . Let $T \subset S$ be a chain. We claim that:

$$(1.8) \quad E_{\text{upp}} = \bigcup_{E \in T} E$$

is an upper bound of T . Indeed, first note that if $\alpha, \beta \in E_{\text{upp}}$ then $\alpha \in E_1 \in T$ and $\beta \in E_2 \in T$. Since every two elements of T are comparable, we know that either $E_1 \leq E_2$ or $E_2 \leq E_1$. Assume that $E_1 \leq E_2$. In this case we have $\alpha, \beta \in E_2$ so $\alpha + \beta, \alpha\beta, 1/\alpha \in E_2$ (where $1/\alpha \in E_2$ only holds if $\alpha \neq 0$). It follows that if $\alpha, \beta \in E_{\text{upp}}$ then $\alpha + \beta, \alpha\beta \in E_{\text{upp}}$ and if $\alpha \neq 0$ then $1/\alpha \in E_{\text{upp}}$, so E_{upp} forms a field. Moreover, $F \subset E_{\text{upp}}$ as long as $T \neq \emptyset$ since $F \subset E$ for any $E \in S$. That is, E_{upp} is an extension of F . Finally, if $\alpha \in E_{\text{upp}}$ then $\alpha \in E$ for some algebraic extension E of F , so α is algebraic over F . That is, every element of E_{upp} is algebraic over F so E_{upp} is an algebraic extension of F . We conclude that $E_{\text{upp}} \in S$, and clearly $E \leq E_{\text{upp}}$ for every $E \in T$ so T has an upper bound.

By Zorn's lemma we conclude that there is a maximal element $\overline{F} \in S$. We claim that $\overline{F}$ is algebraically closed. Indeed, suppose that $\overline{F}$ was not algebraically closed. In that case there would be some non-constant $f(x) \in \overline{F}[x]$ such that $f(x)$ has no zero in $\overline{F}$. However, we know that there exists an algebraic extension of $\overline{F}$, E , such that $f(x)$ has a zero in E . In fact, if we let $\alpha \in E$ be a zero of $f(x)$ then there is no loss of generality to assume that $E = \overline{F}(\alpha)$. We claim that $\overline{F}(\alpha)$ is an element of S , this would then contradict maximality of $\overline{F}$ since $\overline{F} \leq \overline{F}(\alpha)$ but $\overline{F}(\alpha) \neq \overline{F}$ (since $\alpha \notin \overline{F}$). To see that $\overline{F}(\alpha) \in S$, note that we can write:

$$(1.9) \quad f(x) = c_0 + \dots + c_n x^n$$

with $c_0, \dots, c_n \in \overline{F}$ algebraic over F . We note that $F(c_0, \dots, c_n)$ is a finite extension of F , and, as in the argument that $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$, we note that $F(c_0, \dots, c_n)(\alpha)$ is a finite extension of $F(c_0, \dots, c_n)$ from which it follows that $F(c_0, \dots, c_n)(\alpha)$ is a finite extension of F . Since any finite extension is algebraic, we conclude that $\alpha \in F(c_0, \dots, c_n)(\alpha)$ is algebraic over F . It follows that $\overline{F}(\alpha) =$

$\{b_0 + b_1\alpha + \dots + b_{n-1}\alpha^{n-1} \mid b_0, \dots, b_{n-1} \in \overline{F}\}$ is an algebraic extension of F since the algebraic elements of $\overline{F}(\alpha)$ form a subfield of $\overline{F}(\alpha)$, and any element in $\overline{F}(\alpha)$ can be written as a combination of sums and products of algebraic elements over F . That is $\overline{F}(\alpha) \in S$ which contradicts the maximality of $\overline{F}$.

By this contradiction we conclude that $f(x) \in \overline{F}[x]$ must have a zero in $\overline{F}$, so $\overline{F}$ is algebraically closed. ■

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